CPT208 Human-Centric Computing

03. Conceptual Prototyping and Practical Guide

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Today's plan

1. Introduction of prototyping
2. How to use prototyping
3. Common tools in prototyping
4. Fidelity in prototyping

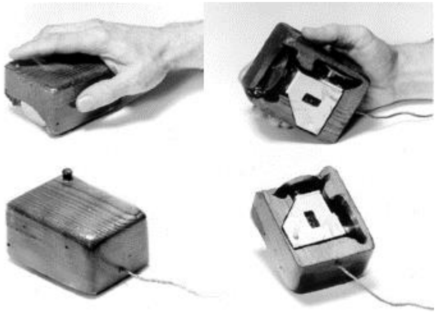
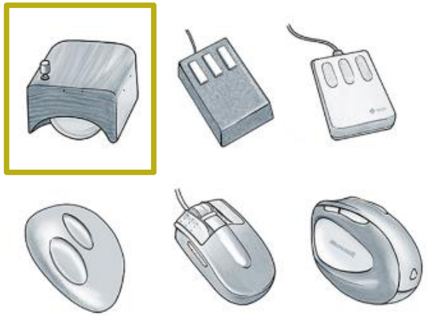
Prototyping

Chapter 12

What is a prototype?

A **prototype** is an early model or representation of a product designed to test an idea or function before committing to full production. It transforms abstract concepts into tangible experiences that can be evaluated, tested, and refined.

In design thinking, prototyping is the bridge between **ideation** and **testing**, where imagination meets execution.

What is a prototype?

The Oxford definition of a prototype is:

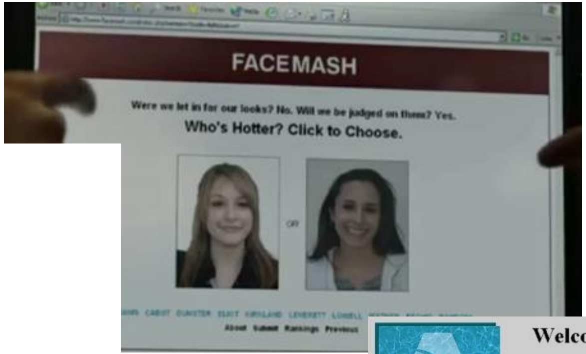
- "A first, typical or preliminary model of something, especially a machine, from which other forms are developed or copied"

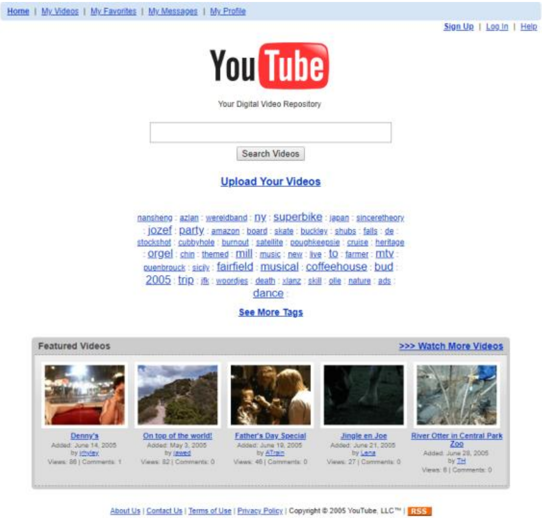
The word originates from the Greek ***prōtotupos***

- meaning "first example"

What is a prototype? - **Physical**

What is a prototype? - **Digital**





What are these famous websites?

Characteristics of Prototyping

Goal-Oriented

A good prototype is built to answer a specific question. (e.g., "Will a user intuitively know how to find the 'submit' button?"). It prevents aimless design.

Disposable & Flexible

The true value lies in the knowledge gained, not the artifact itself. Teams must be willing to throw a prototype away if it doesn't serve the user's ultimate needs.

Iterative

It is not a one-and-done process. Prototypes evolve continuously through cyclical phases of creation, user testing, and refinement based on direct feedback.

CORE QUESTION

Does this idea solve the problem we defined?

By creating a prototype, teams move from thinking to doing, transforming hypotheses into physical or digital experiences that can be evaluated.

What is a prototype? **- Beyond**

- It is a **tool for thinking** and learning, not just a stepping stone to the final product.
- It actively bridges the gap between abstract requirements and tangible, testable experiences.
- It reduces ambiguity for both designers and stakeholders early in the process.

> *"Fail fast to succeed sooner."*

Why prototype?

- **Evaluation and feedback** are central to interaction design
- Stakeholders **can see, hold, interact** with a prototype more easily than a document or a drawing
- Team members can **communicate effectively**
- You can **test out ideas** for yourself
- It encourages **reflection**: very important aspect of design
- Prototypes answer questions, and support designers in choosing between alternatives

Why prototype? - The Mindset

In software development, building the wrong thing perfectly is the most expensive mistake you can make. Prototyping shifts our mindset from **"I know what works"** to **"Let's test what works."**

- **Failing fast and cheaply**
- **Lowering the stakes**
- **Separating ego from design**

The "Notes App" Analogy

Imagine you need to send a highly sensitive or risky text message (e.g., asking for an extension, or texting a crush).

You rarely type it directly into the chat and hit send. Instead, you draft it in your Notes app. You read it back. You tweak the tone. You might even show your roommate and ask, *"Does this sound weird?"*

That is prototyping!
You are testing the interaction in a safe environment before deploying it to the real world.

Why prototype? - The Tool

The "Dinner Dilemma" Analogy

Have you ever asked a group of friends, *"What do you want for dinner?"* usually, the answer is a blank stare or *"I don't know."*

But if you propose, *"Let's get spicy tacos,"* suddenly everyone has an opinion: *"No, my stomach hurts, let's get soup."* or *"No, we had Mexican yesterday."*

People struggle to invent solutions from scratch, but they are brilliant at reacting to a concrete proposition.

Human beings are highly visual and reactive. We often don't know what we want until we see exactly what we **don't** want.

- **Eliciting actionable feedback**
- **Revealing hidden requirements**
- **Bypassing imagination**

How do we use Prototype

Prototyping extends beyond simply building a product. It serves three vital cognitive and social functions in the design process.

Understanding & Ideation

Exploring complex problem spaces, uncovering technical or user constraints, and sparking innovative ideas through the act of making.

Communication

Providing a tangible, shared artifact that aligns diverse stakeholders—bridging the gap between users, developers, and business leaders.

Task & Reflection

Evaluating specific designs against real-world use cases, allowing teams to test hypotheses and reflect on empirical user data.

Prototype - **Understanding & Ideation**

Prototyping is thinking through making. It helps us uncover hidden requirements and sparks new ideas long before writing a single line of production code.

THINGS TO CONSIDER

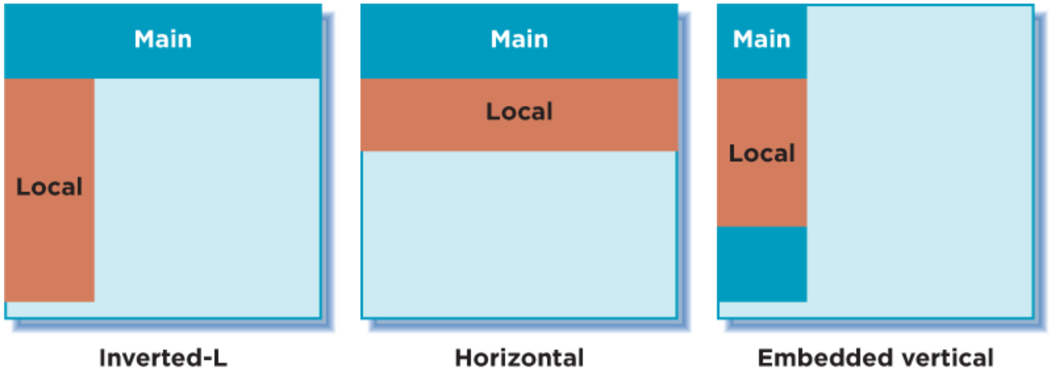
- **Embrace Ambiguity:** Use early prototypes to ask questions, not just to provide definitive answers.
- **Keep Flexible:** Focus details too early can stifle divergent thinking. Stick to sketches or simple wireframes initially.
- **Generate Variants:** Build multiple, distinctly different versions of a feature to thoroughly explore the solution space.

Understanding & Ideation - **alternatives**

"All roads lead to Rome"

→ How can you be sure that the one you choose is the right one for your design?

→ Don't be sure, make many of them and try!



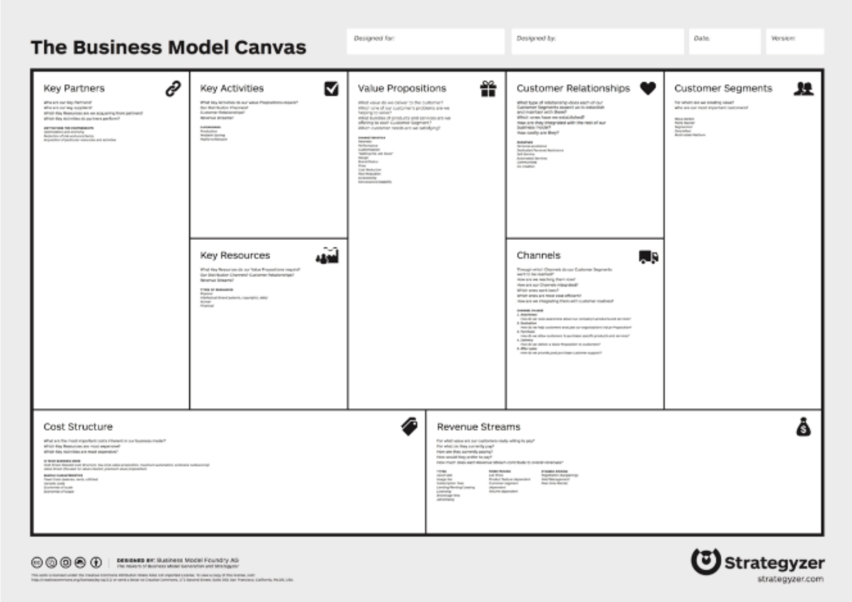
Understanding & Ideation - **alternatives**

Understanding & Ideation - **strategy**

What are pros and cons?

➔ How would you compete with other designs?

➔ **Intangible -> tangible**

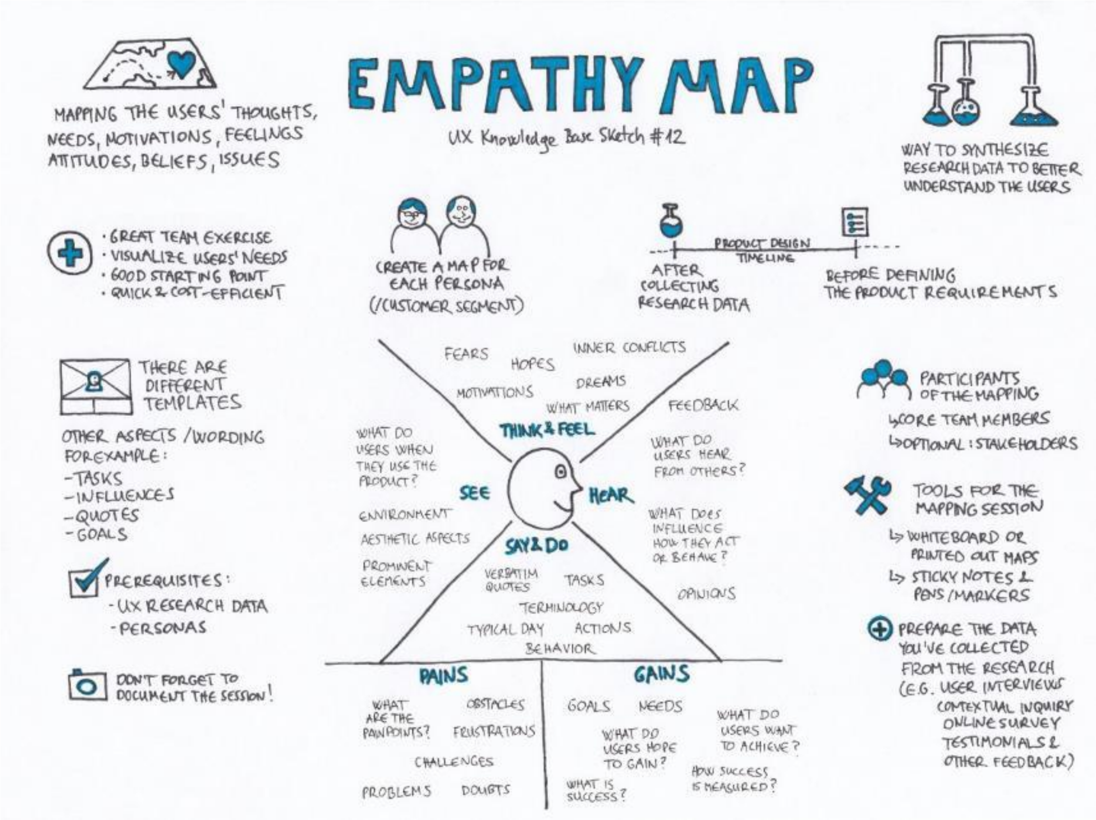


Understanding & Ideation – **User-centred Process**

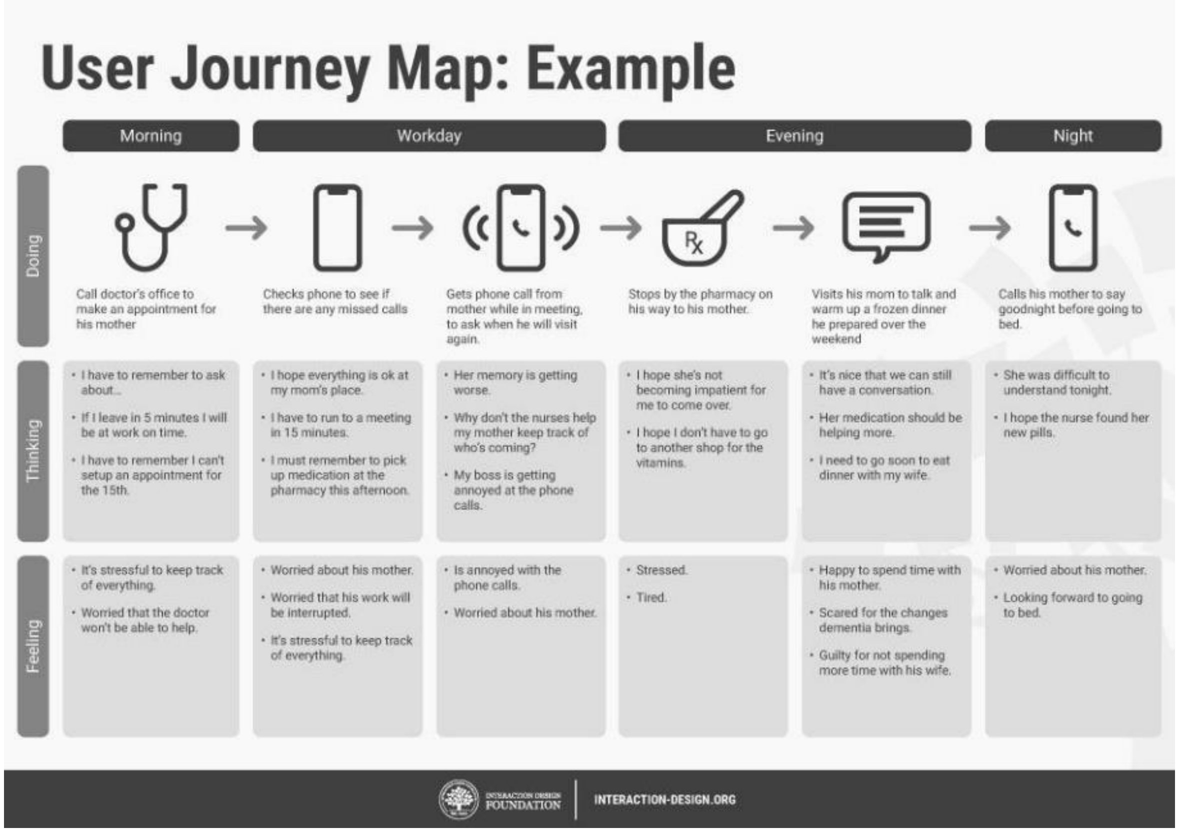
User-centred design

- Don’t make false assumptions
- Prototype could build **empathy** between designers and users with concrete ideas
- Useful tools:
  - Empathy map
  - User journey map

User-centred Process **- Empathy map**



User-centred Process - **User journey map**



User journey map – **for a real journey**

FINLAND

SEPTEMBER (2019)

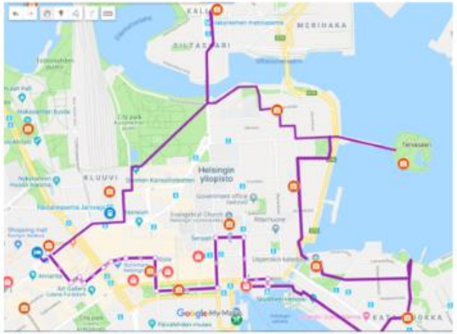
DAY 6 (HELSINKI) – SUNDAY 15 SEPTEMBER

MORNING
- Breakfast buffet at hotel
- 8 - 8:30am at latest Walk to Hakaniemi Market Hall via City Park
- 9am - Hakaniemi Market Hall
- 9:45am walk around Kruununhaka neighbourhood
- Walk past the Burgher's Home (oldest wooden building in Helsinki)
- 10:15am Walk around Tervasari Island
- 10:45am - Walk to Uspenski Cathedral
- 11:15am – Walk around Katajanokka areas

AFTERNOON
- 12pm lunch at the Old Market Hall
- 1:30pm Ferry to Suomenlinna Island
- 5pm return to Helsinki
- 5:15pm Helsinki Cathedral (outside only)
- Walk through Espanadi Park

EVENING
- Shopping in Esplanadi area
- Dinner at Zetor restaurant

Purple Route – Total Walking = approx. 12km.



Accommodation: *Scandic Hotel Simonkenttä (pre-paid)*

Prototype - **Communication**

Prototypes act as powerful "boundary objects." They create a shared visual language that helps non-technical stakeholders and developers align on the product vision instantly.

THINGS TO CONSIDER

- **Tailor to the Audience:** Executives may need a high-level interactive flow, while developers need functional logic and edge cases.
- **Focus on Narrative:** Use the prototype to tell a story about the user journey, rather than just clicking through isolated screens.
- **Direct the Feedback:** Explicitly tell stakeholders what you are testing (e.g., "Ignore the colors; focus on the checkout steps") to prevent unhelpful critiques.

Communication - **Consensus**

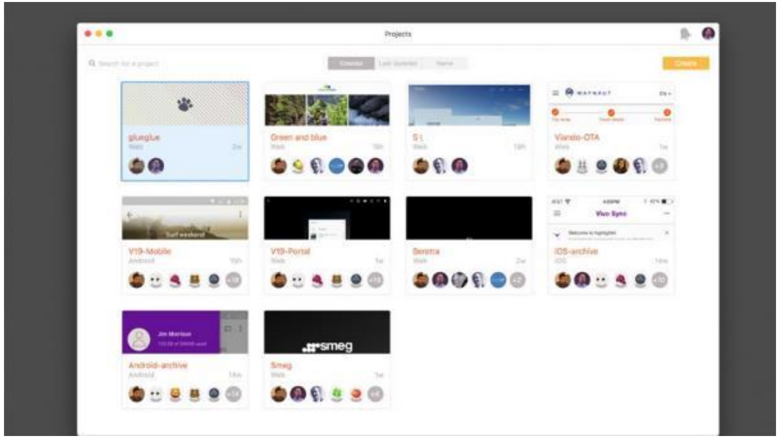
Communication - **Stakeholders**

Imagine how would you present the prototype with these stakeholders:

- ➔ Investors (Highlights/potentials)
- ➔ Factory (Materials, scales)
- ➔ Users
  - Elderly (caring?)
  - Child (warmth?)
  - Male or female (Cost or privilege?)

Communication - **Focus**

Prototype could save efforts in determining **style, size, colour, and interactions** through **observable objects** and **peer discussion**



Prototype - **Test & Reflection**

Prototypes allow us to evaluate specific tasks in context. They ground our design decisions in empirical evidence rather than subjective assumptions.

THINGS TO CONSIDER

- **Define Clear Metrics:** Before testing, establish what success looks like (e.g., time to complete a task, error rate, task success rate).
- **Observe Behavior:** Pay close attention to what users do, not just what they say. Watch for hesitations, misclicks, and workarounds.
- **Reflect and Iterate:** Systematically synthesize the data gathered to inform the next iteration of the design. Testing without reflection is wasted effort.

Test & Reflection – **Hypotheses and assumptions**

When we begin the design, we have to make assumptions no matter how hard we try to understand users, for example:

- ➔ Users could find the way to specific functions
- ➔ The information displayed is straightforward
- ➔ Users could understand the UI and texts
- ➔ Users‘ needs could be fully satisfied within this page

The only way to find the answer is to let them play with a mock system, like you need to **practice before the presentation**

Test & Reflection - **Others' comments**

The test and reflection are not only about users, but also **within the design group**

The prototype is an opportunity for you to **present and promote the idea to peers**, and reflect on the availability and values from different perspectives

Achieving Prototyping Goals

To effectively leverage these three purposes, maintain these overarching principles:

Align Purpose and Project Stage

Ensure the prototype‘s primary goal (ideation vs. final validation) matches your current phase. Don’t use a polished prototype when you are still trying to understand basic user needs.

Avoid the "Polish Trap"

Resist the urge to spend hours perfecting visual aesthetics if your primary goal is to test a structural workflow or communicate a basic concept. Time is better spent iterating.

Foster a Learning Culture

Treat every prototype as a scientific experiment designed to generate insights. A prototype that proves an idea is flawed is just as valuable as one that proves an idea is successful.

Fidelity in prototyping

Chapter 12

What is fidelity?

The fidelity of a prototype **refers to how it conveys the look-and-feel of the final product** (basically, its level of detail and realism).

Select the right level of fidelity in prototyping is the key to the success of design process.

➔ Lower fidelity may be useless, higher fidelity may be time- and money-consuming

Fidelity –
A Measurement, Not a Classification

We often hear the terms "Low-Fi" and "High-Fi" thrown around as if they are strict, separate categories. In reality, fidelity is a **measurement system.**

Think of it like a volume dial on a speaker or the resolution setting on a YouTube video (144p vs. 4K). It measures the degree of exactness or detail with which a prototype matches the final, intended product.

0%	**30%**	**70%**	**99%**
Abstract Idea / Whiteboard scribble	Paper Wireframes / Basic layouts	Clickable digital screens / Fake data	Coded frontend / Live API data

"How much fidelity do we need to buy the learning we seek?"

Why Did This System Emerge?

Historically (in the 1980s/90s software era), teams used the "Waterfall" method. They would design entirely on paper, and then build 100% of the software before testing it. This was disastrous.

The Cost Tension: Developers realized that making changes at the "100% finished" stage was incredibly expensive and soul-crushing.

The Need for a Vocabulary: As Agile and Lean methodologies emerged, teams needed a way to negotiate with stakeholders. They needed a shared language to say, *"We built a 20% version to test the waters."*

Validating Assumptions: The system emerged to break the massive risk of software development into smaller, measurable, and testable milestones.

Fidelity Scale - **Strategic Value**

Treating fidelity as a variable dial (rather than a fixed state) gives designers a powerful strategic advantage:

Resource Allocation

It prevents over-engineering. Why spend 40 hours coding a high-fidelity dropdown menu if a 10-minute sketch can prove that users don't even want a dropdown menu in the first place?

Directing Feedback

Lower fidelity forces stakeholders to focus on structure and flow. If you show a 90% fidelity mockup too early, executives will argue about the shade of blue instead of the core user journey.

Multi-Dimensional Testing

It teaches us that fidelity isn't just visual! You can have a prototype that looks terrible (low visual fidelity) but functions perfectly with real data (high functional/data fidelity).

Quick Start - **The Quick & Dirty Arsenal**

Sticky Notes

The universal tool for brainstorming and affinity mapping. Highly flexible, allowing teams to quickly group ideas, map initial user flows, and cluster insights during ideation phases.

Paper & Pen

Offers a zero-barrier to entry for sketching initial interfaces. It prevents designers from getting distracted by pixels or colors, forcing a focus purely on layout and core functionality.

Index Cards

Perfect for card sorting exercises and modular screen flows. They help structure information architecture by allowing users to physically organize content categories.

Storyboards

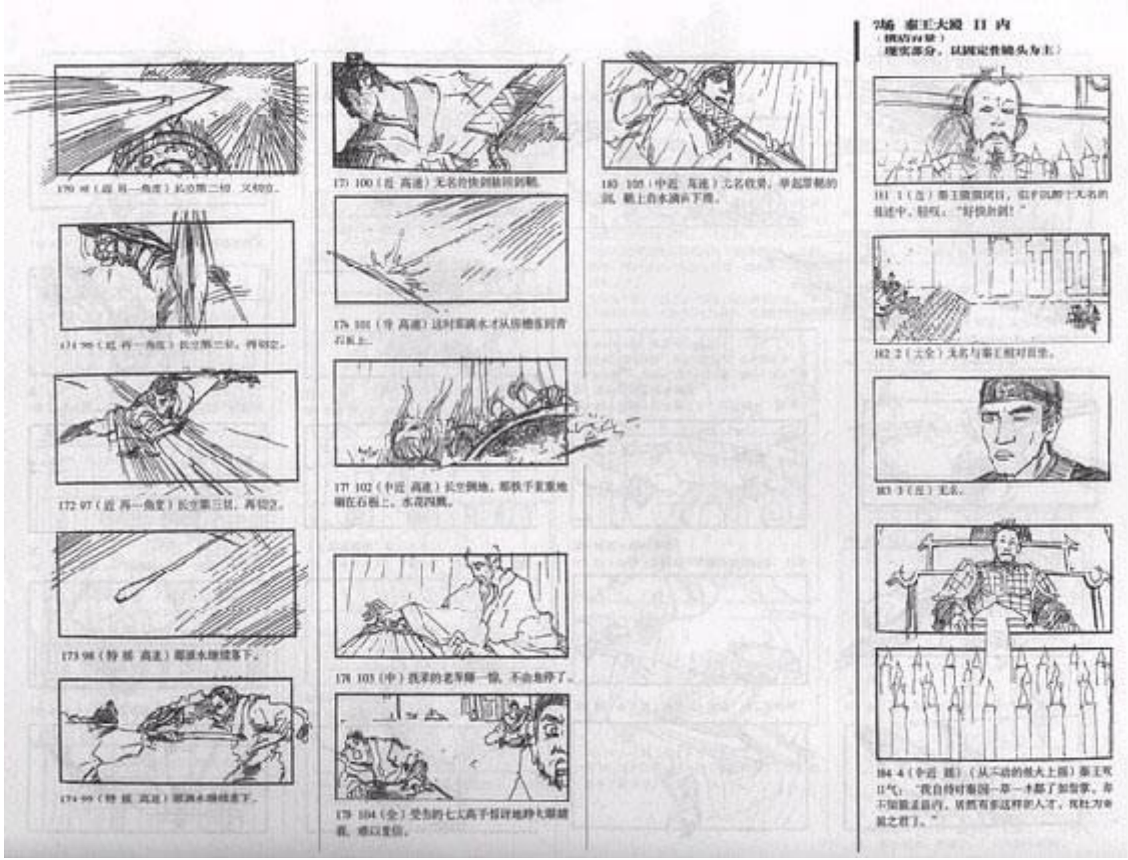
Visualizing the Context

Storyboards map out a scenario frame-by-frame, similar to a comic book. They are crucial for showing how a user interacts with your solution in the real world, rather than just looking at the interface.

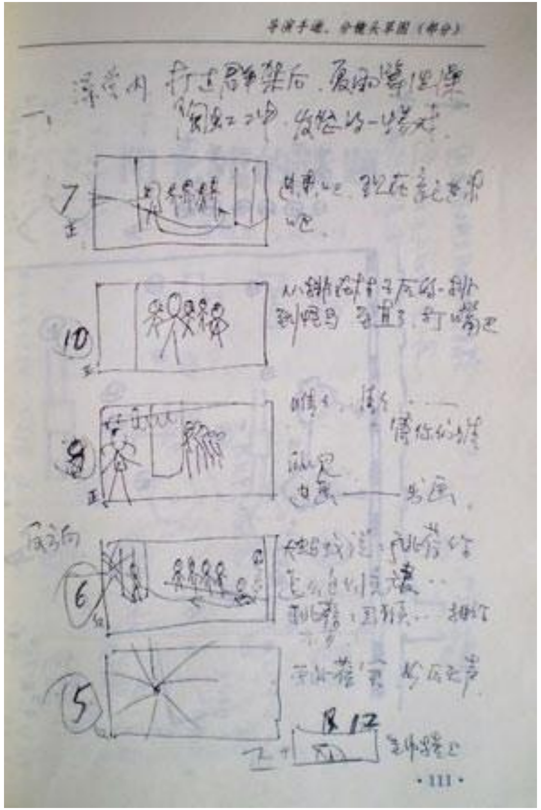
Primary Benefit: By placing the product in a realistic environment, storyboarding builds deep empathy and highlights environmental constraints that standard wireframes miss.

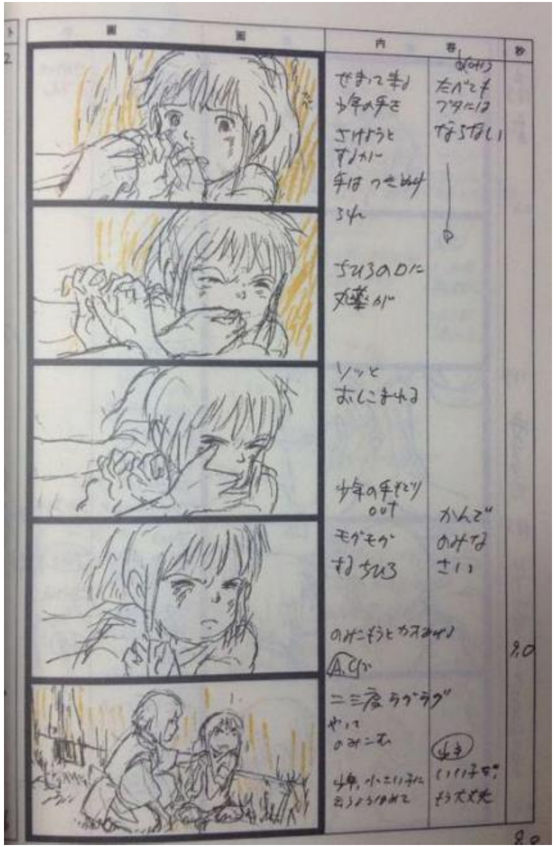
Storyboards

Storyboards - **no perfection**



Storyboards - **no perfection**

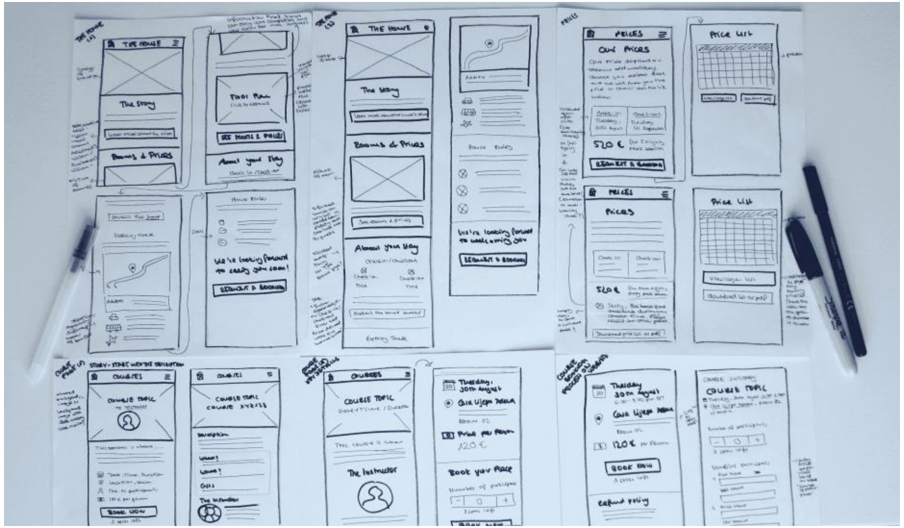




Tools - **Sketching**

Sketching is important to low-fidelity prototyping

Same with the storyboard - **Don't be inhibited about drawing ability**. Practice simple symbols with pencil and paper



Tools - **Index Cards**

Index cards (3 X 5 inches)

Each card represents **one screen**

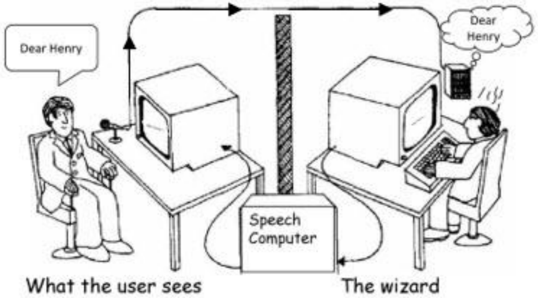
Often used in

- → **Website development**
- → **Application**
- → **MiniProgram**

Tools - **‘Wizard-of-Oz’**

- The user thinks they are interacting with a computer, but a **developer is responding** to output rather than the system.
- Usually **done early** in design to understand users’ expectations
- What is ’wrong’ with this approach?
  - It is “Artificial Intelligence”

Wizard of Oz testing – The listening type writer IBM 1984



‘Wizard-of-Oz’ prototyping

Understanding **Trade-offs**

➔ Prototyping is inherently a compromise. You cannot achieve absolute high fidelity, zero cost, and immediate delivery simultaneously. Recognizing these tensions is key to effective product strategy.

- A paper sketch is fast to create but yields little interaction data. A coded prototype yields rich data but limits your ability to rapidly pivot if the core idea fails.
- You must decide whether to test a shallow version of the entire user journey (horizontal prototype) or a highly detailed version of a single specific feature (vertical prototype).
- Internal executives often require polished, high-fidelity visuals to secure funding, whereas end-users actually provide better structural feedback on unpolished wireframes where they aren't distracted by colors.

How to **Navigate** Compromises

Define the Core Hypothesis

Before selecting a tool, ask one critical question: *"What is the single most important thing we need to learn right now?"*

Once you define the core unknown, choose the cheapest, fastest tool capable of answering exactly that question, and nothing more. Do not overbuild.

Embrace Disposable Artifacts

A prototype's sole purpose is to buy knowledge. It is not the final product.

If a prototype takes so long to build that you find yourself defending it against user critique, your fidelity was too high. Build to learn, and be prepared to throw it away.

"

If a picture is worth 1000 words, a prototype is worth 1000 meetings.

"

— IDEO / TOM & DAVID KELLEY

Any Questions?

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